

Set

$$|A|f : A \rightarrow B|A| = |B|A, B\mathbf{Set}$$

$$|A \sqcup B| = |A| + |B|$$

$$|A \times B| = |A| \times |B|$$

$$|A^B| = |A|^{|B|}B \rightarrow A$$

$$A \times (B \sqcup C) \cong (A \times B) \sqcup (A \times C)$$

$$f:X\rightarrow Yf\in fX=f\in Y=f\in$$

$$X=\{a,b,c\}Y=\{a,e\}$$

$$X=\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=\begin{pmatrix}a\\b\\c\end{pmatrix},Y=\begin{pmatrix}y_1\\y_2\end{pmatrix}=\begin{pmatrix}a\\e\end{pmatrix}$$

$$XY(x,y)\in X\times Y\otimes x\otimes y\in ZZ$$

$$\begin{array}{l} \otimes:X\times Y\rightarrow Z=X\times Y\\ (x,y)\mapsto x\otimes y=(x,y) \end{array}$$

$$x\otimes y=(x,y)X\otimes Y=X\times YXYXYX\otimes Y$$

$$\begin{aligned}
 X\otimes Y &= \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \otimes \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \otimes \begin{pmatrix} a \\ e \end{pmatrix} \\
 &= \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \otimes Y = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \otimes Y \\
 &= \begin{pmatrix} x_1 \otimes Y \\ x_2 \otimes Y \\ x_3 \otimes Y \end{pmatrix} = \begin{pmatrix} a \otimes Y \\ b \otimes Y \\ c \otimes Y \end{pmatrix} \\
 &= \begin{pmatrix} x_1 \otimes y_1 \\ x_1 \otimes y_2 \\ x_2 \otimes y_1 \\ x_2 \otimes y_2 \\ x_3 \otimes y_1 \\ x_3 \otimes y_2 \end{pmatrix} = \begin{pmatrix} a \otimes a \\ a \otimes e \\ b \otimes a \\ b \otimes e \\ c \otimes a \\ c \otimes e \end{pmatrix} = \begin{pmatrix} (a,a) \\ (a,e) \\ (b,a) \\ (b,e) \\ (c,a) \\ (c,e) \end{pmatrix} \\
 &= \{(x_i,y_j) | 1 \leq i \leq 3, 1 \leq j \leq 2\}
 \end{aligned}$$

$$|X\otimes Y|=|X\times Y|=|X|\times|Y|$$

$$xX=\{1,x,x^2,\cdots\}\mathbb{R}X\mathbb{R}$$

$$p(x)=3x^3-2x+5$$

$$pX\mathbb{R}$$

$$p:X\rightarrow \mathbb{R}$$

$$x^i\mapsto a_i$$

$$p(x)=a_0+a_1x+a_2x^2+\cdots a_i\neq 0X=\{1,x,x^2,\cdots\}\mathbb{R}\mathbb{R}$$

$$\underbrace{X=\begin{pmatrix}1\\x\\x^2\\x^3\\\vdots\end{pmatrix}}\rightarrow \underbrace{p(x)=\begin{pmatrix}5\\-2x\\0\\3x^3\\\vdots\end{pmatrix}}\rightarrow \underbrace{p=\begin{pmatrix}5\\-2\\0\\3\\\vdots\end{pmatrix}}$$

$$pX\mathbb{R}\mathbb{R}$$

$$\mathbb{R}^X=\{p:X\rightarrow\mathbb{R}\}$$

$$|\mathbb{R}^X|=|\mathbb{R}|^{|X|}$$

$$\mathbb{R}^X=\left\{\left[\begin{matrix}a_1\\a_2\\\vdots\end{matrix}\right],a_i\in\mathbb{R}\right\}$$

$$\mathbb{R}[X]\mathbb{R}^XX=\{1,x,x^2,\cdots\}$$

$$X\in I\in |I|=|X|i\in Ix_i\in X$$

$$\begin{array}{l} p:X\rightarrow\mathbb{Z} \\ x_i\mapsto p(x_i)=n_i \end{array}$$

$$\mathbb{Z}^X=\{p:X\rightarrow\mathbb{Z}\}=(X,\mathbb{Z})=(X,\mathbb{Z})$$

$$(-,-)(-,-)p\in\mathbb{Z}^XX$$

$$p(x_i, n_i)$$

$$p_X = \{(x_i, n_i)\}_{i \in I}$$

xz

$$\underbrace{X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_{\text{input}} \rightarrow \underbrace{p_X = \begin{pmatrix} n_1 x_1 \\ n_2 x_2 \\ n_3 x_3 \end{pmatrix}}_{\text{intermediate}} \rightarrow \underbrace{p = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}}_{\text{output}}$$

$$\mathbb{Z}^X = \left\{ \begin{bmatrix} n_1 \\ n_2 \\ \vdots \end{bmatrix}, n_i \in \mathbb{Z} \right\}$$

$$\dim \mathbb{Z}^X = |X|$$

$$n_i = 0n_ix_in_i \neq 0\mathbb{Z}[X]\mathbb{Z}^X$$

$$\mathbb{Z}[X] \times \mathbb{Z}[X] \rightarrow \mathbb{Z}[X]$$

$$\left(\begin{bmatrix} m_1 \\ m_2 \\ \vdots \end{bmatrix}, \begin{bmatrix} n_1 \\ n_2 \\ \vdots \end{bmatrix} \right) \mapsto \begin{bmatrix} m_1 \\ m_2 \\ \vdots \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} m_1 + n_1 \\ m_2 + n_2 \\ \vdots \end{bmatrix}$$

$$m_in_jm_i+n_j$$

$$X\mathbb{Z}[X]\wedge 0\wedge \wedge [X]0$$

$$\mathbb{Z}[X]\mathbb{Z}=(\mathbb{Z},+,0)\in \mathbb{Z}\mathbb{Z}[X]$$

$$\begin{array}{l} F:~\rightarrow \\ X\mapsto FX=\mathbb{Z}[X] \\ Y\mapsto FY=\mathbb{Z}[Y] \end{array}$$

$$X\otimes Y=X\times YF(X\otimes Y)=\mathbb{Z}[X\otimes Y]FX=\mathbb{Z}[X]FY=\mathbb{Z}[Y]$$

$$\dim \mathbb{Z}^{X\otimes Y} = \dim \mathbb{Z}[X\otimes Y] = |X\otimes Y| = |X|\times |Y| = \dim \mathbb{Z}^X \times \dim \mathbb{Z}^Y$$

$$X\otimes Y = X\times Y \xrightarrow[\dim \mathbb{Z}[X\otimes Y]]{F} \mathbb{Z}[X\otimes Y] \xrightarrow[\dim X]{\dim Y} X\otimes Y = |X|\times |Y|$$

$$F(X\otimes Y)=F\begin{pmatrix}a\otimes a\\a\otimes e\\b\otimes a\\b\otimes e\\c\otimes a\\c\otimes e\end{pmatrix}$$

$$\underbrace{X \otimes Y = \begin{pmatrix} a \otimes a \\ a \otimes e \\ b \otimes a \\ b \otimes e \\ c \otimes a \\ c \otimes e \end{pmatrix}} \rightarrow \underbrace{p_{X \otimes Y} = \begin{pmatrix} n_{a \otimes a}(a \otimes a) \\ n_{a \otimes e}(a \otimes e) \\ n_{b \otimes a}(b \otimes a) \\ n_{b \otimes e}(b \otimes e) \\ n_{c \otimes a}(c \otimes a) \\ n_{c \otimes e}(c \otimes e) \end{pmatrix}} \rightarrow \underbrace{p = \begin{pmatrix} n_{a \otimes a} \\ n_{a \otimes e} \\ n_{b \otimes a} \\ n_{b \otimes e} \\ n_{c \otimes a} \\ n_{c \otimes e} \end{pmatrix}}$$

$$p|X| \times |Y|$$

$$F:\rightarrow$$

$$X \mapsto FX = \mathbb{Z}[X]$$

$$Y \mapsto FY = \mathbb{Z}[Y]$$

$$X \otimes Y \mapsto F(X \otimes Y) = \mathbb{Z}[X] \otimes \mathbb{Z}[Y]$$

$$\mathbb{R}[x]\mathbb{R}[y]$$

$$\begin{cases} p_1(x)=3x+1 \\ q_1(y)=-2y \end{cases} \begin{cases} p_2(x)=-6x-2 \\ q_2(y)=y \end{cases}$$

$$(p_1,q_1)=p_1\otimes q_1\in\mathbb{R}[x]\otimes\mathbb{R}[y](p_2,q_2)=p_1\otimes q_1\in\mathbb{R}[x]\otimes\mathbb{R}[y](p_1,q_1)\neq (p_2,q_2)\mathbb{R}[x,y]$$

$$p_1(x)\cdot q_1(y)=p_2(x)\cdot q_2(y)=-6xy-2y$$

$$\begin{array}{lll} (p_1,q_1)(p_2,q_2) & \mathbb{R}[x]\otimes\mathbb{R}[y] & \mathbb{R}[x]\otimes\mathbb{R}[y] \\ (p_1,q_1)\sim (p_2,q_2) & & p_1(x)\cdot q_1(y)=p_2(x)\cdot q_2(y) \end{array}$$

$$\mathbb{R}[x]\otimes\mathbb{R}[y]=\mathbb{R}[x]\otimes\mathbb{R}[y]/\sim$$

$$(p_1,q_1)\sim (p_2,q_2)\Longleftrightarrow p_1\cdot q_1=p_2\cdot q_2\Longleftrightarrow p_1\cdot q_1-p_2\cdot q_2=0$$

$$X=\{a,b,c\}XWW[aab]+[cb]=[aabc b][\,](W,+,[\,])$$

$$X\{a,a^{-1},b,b^{-1},c,c^{-1}\}(FX,+,[\,])[xx^{-1}]=[x^{-1}x]=[\,](FX,+,[\,])X$$

$$\begin{array}{l} \mathbb{Z} \quad n \in \mathbb{Z} \, n\mathbb{Z} = \{nk|k \in \mathbb{Z}\} \, (n) \quad a-b \in n\mathbb{Z} \Leftrightarrow a \sim_n b \Leftrightarrow a \equiv b \, (\text{mod } n) \\ \mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/\sim_n = \{i+n\mathbb{Z}\} = \{[0]_n,[1]_n,\cdots,[n-1]_n\} \, \mathbb{Z}_n = \{0,1,\cdots,n-1\} \\ |\mathbb{Z}_n| = n\mathbb{Z}_n = \mathbb{Z}/(n)[a] \cdot [b] = [a \cdot b] \end{array}$$

$$\begin{aligned}\mathbb{Z}_m\otimes\mathbb{Z}_n&=\left([i]_m\right)\otimes\left([j]_n\right)\\&=\left([i]_m\otimes[j]_n\right)=\left((i+m\mathbb{Z})\otimes(j+n\mathbb{Z})\right)\end{aligned}$$

$$\begin{array}{l} x\otimes y=(x,y)\,(i+m\mathbb{Z})\otimes(j+n\mathbb{Z})\,(i+m\mathbb{Z},j+n\mathbb{Z})\,\mathbb{Z}_m\mathbb{Z}_n\,\mathbb{Z}_m\otimes\mathbb{Z}_n\,\,o\in\mathbb{Z}^+\\ \mathbb{Z}_m\otimes\mathbb{Z}_n\simeq\mathbb{Z}_o(i+m\mathbb{Z})\otimes(j+n\mathbb{Z})k+o\mathbb{Z} \end{array}$$

$$n=1\mathbb{Z}_1=\mathbb{Z}/1\mathbb{Z}=\{0\}$$

$$\mathbb{Z}_m\otimes\mathbb{Z}_1=\mathbb{Z}_m\otimes\{0\}=\left((i+m\mathbb{Z})\otimes 0\right)\simeq\left(i+m\mathbb{Z}\right)=\mathbb{Z}_m$$

$$\mathbb{Z}_m\otimes\mathbb{Z}_{mn}=\left((i+m\mathbb{Z})\otimes(i+mn\mathbb{Z})\right)\simeq\left(i+m\mathbb{Z}\right)=\mathbb{Z}_m$$

$$\mathbb{Z}_m\otimes\mathbb{Z}_n\simeq\mathbb{Z}_{(m,n)}$$

$$\begin{array}{c} (A\oplus B)\times C\overset{\phi}{\rightarrow}(A\oplus B)\otimes C\\ (A\otimes C)\oplus(B\otimes C)\overset{f}{\rightarrow} \end{array}$$

$$(V\otimes -)\dashv \operatorname{Hom}(V,-)$$

$$\operatorname{Hom}(V\otimes W,U)\cong \operatorname{Hom}(V,\operatorname{Hom}(W,U))$$

$$B:V\times W\rightarrow U\tilde{B}:V\otimes W\rightarrow U$$

$$V\times W\overset{B}{\underset{\cong}{\xrightarrow{\quad}}}V\otimes W$$